



Incorporating 2nd Moment Differential Chromatic Refraction Effects into the Metrics Analysis Framework for LSST

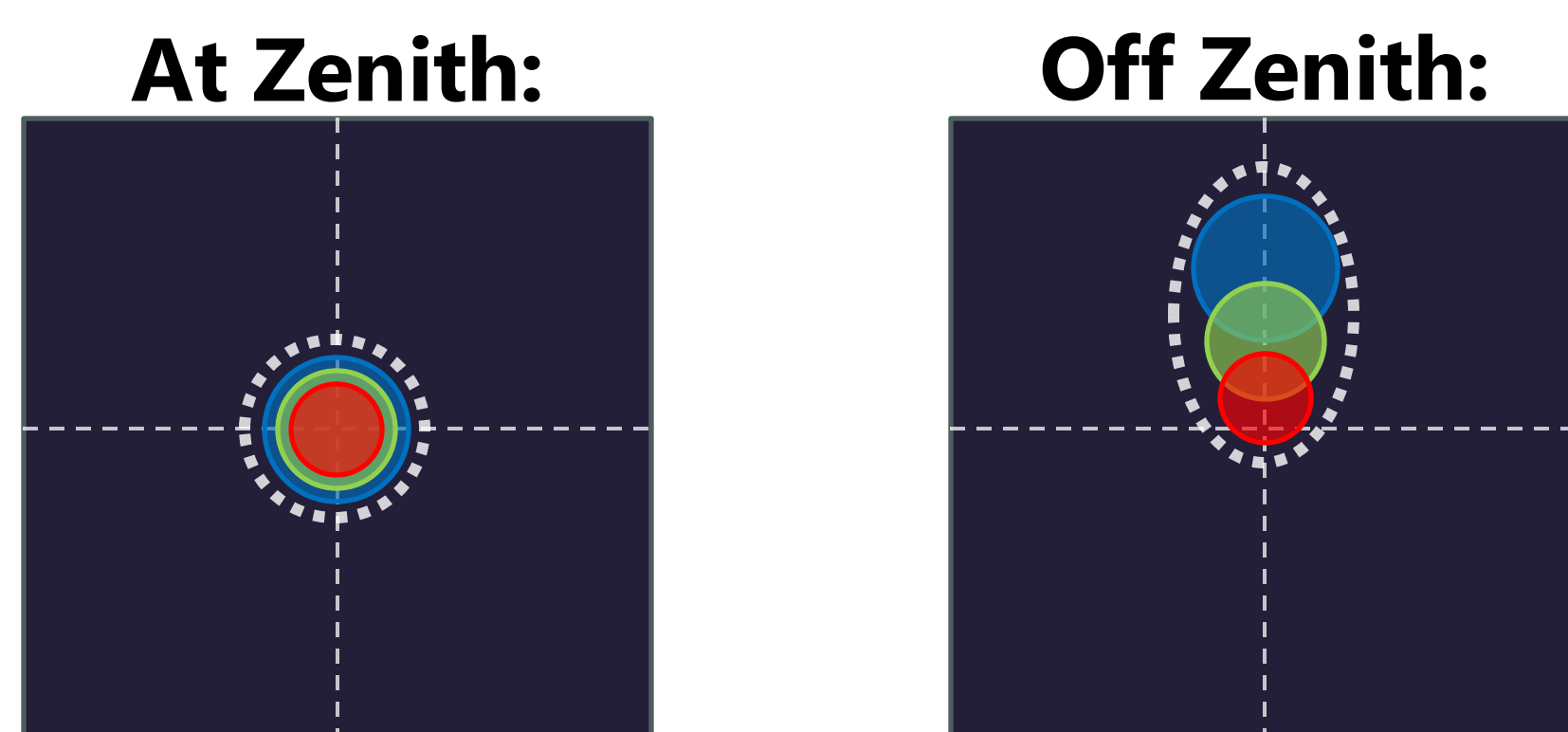


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Differential Chromatic Refraction

- LSST will observe sources over a wide range of zenith angles (**Fig. 1**).
- Light entering the atmosphere at an angle is refracted by an amount that depends on entrance angle and wavelength (**Fig. 2**).
- For broad filters, this refraction results in an increase of the second-moment of the PSF along the zenith direction:



An exaggerated depiction of the DCR effect.

- This effect, called differential chromatic refraction (DCR), decreases signal-to-noise ratio and impacts PSF shapes.
- Second-moment DCR impacts are not yet incorporated into the LSST metric analysis framework (MAF).

OpSim visits with cosmology cuts

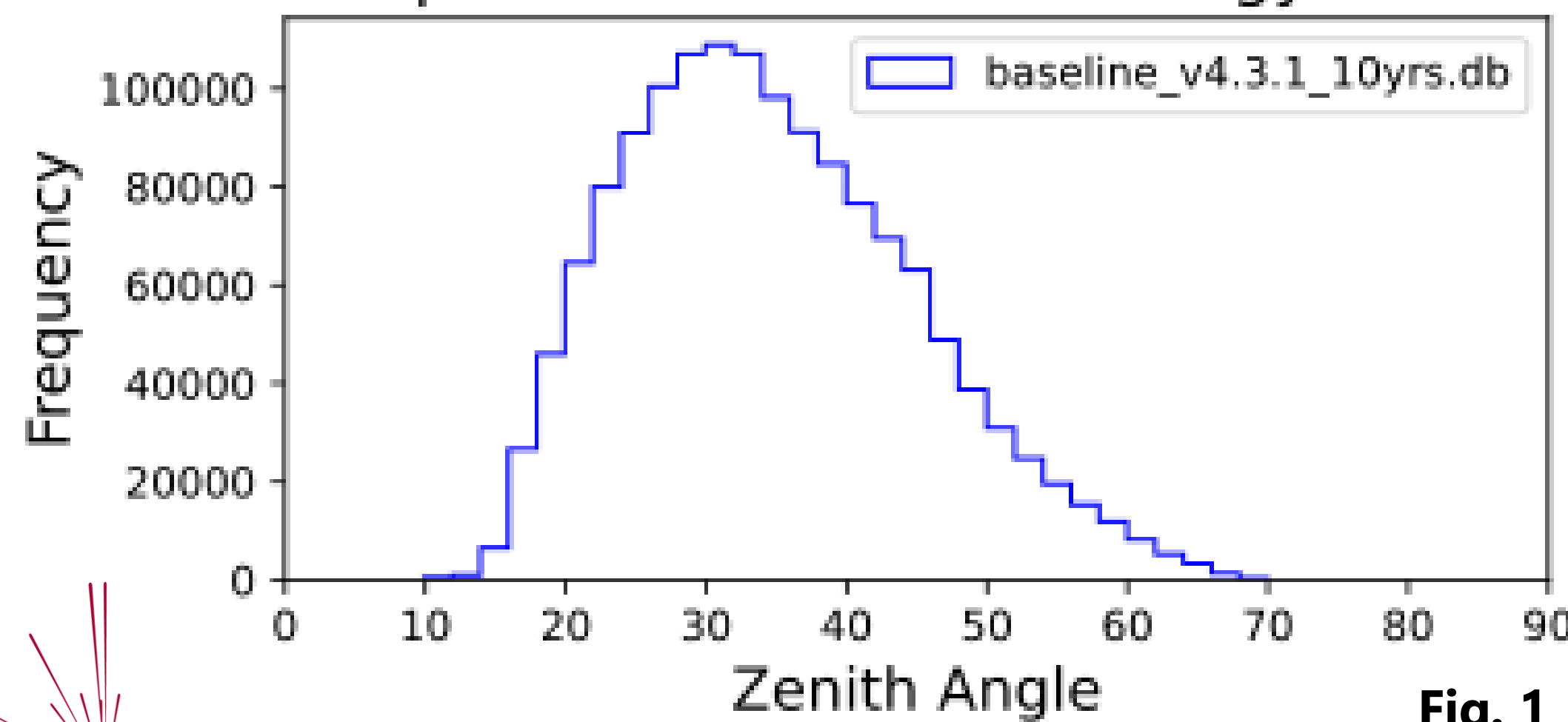


Fig. 1

Estimation of ellipticity due to DCR

- Across a particular band, the distribution of refraction angles, R , of source photons depends on **zenith angle** and the **spectral energy distribution (SED)** of the source (**Fig. 3**).
- Mean variance of refraction angle, $\langle \sigma_R^2 \rangle$, is estimated using 1000 simulated galaxy SEDs and used to estimate PSF ellipticity due to DCR in each band:

$$g_{DCR} = \frac{I_{xx} - I_{yy}}{I_{xx} + I_{yy} + 2\sqrt{I_{xx}I_{yy}}}$$

$$I_{xx} = \left(\frac{\text{FWHM}_{\text{PSF}}}{2.35}\right)^2 \quad I_{yy} = \left(\frac{\text{FWHM}_{\text{PSF}}}{2.35}\right)^2 + \langle \sigma_R^2 \rangle$$

- Fractional increase in PSF size due to DCR as a function of zenith angle is shown in **Fig. 5**.

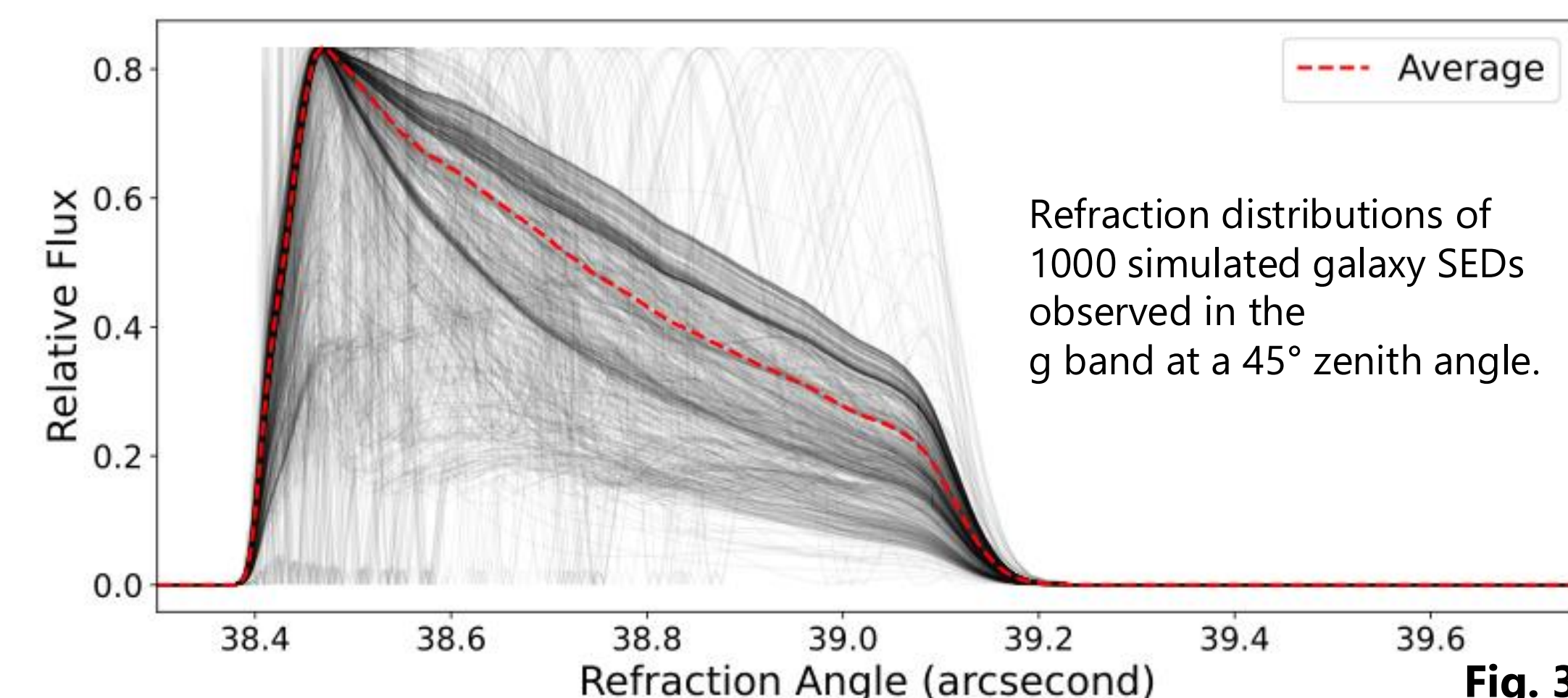
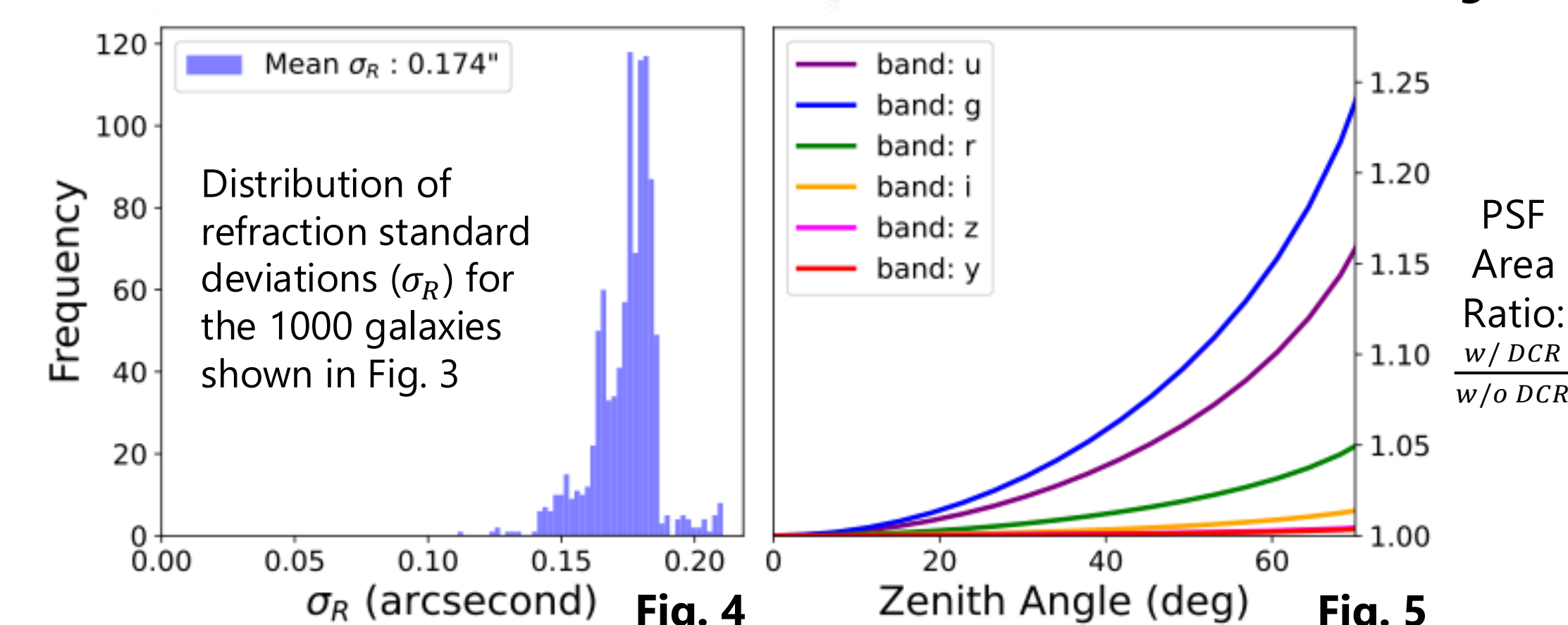


Fig. 3



In progress: a new MAF stacker

- We are currently implementing a new stacker object for MAF that will quantify impacts of DCR on PSF 2nd moments on OpSim visits, enabling more robust analyses of LSST observing strategies.
- Example of the type of plot that could be quickly made with the new stacker, showing maps of the ellipticity due to DCR at a certain percentile of all visits at that location:

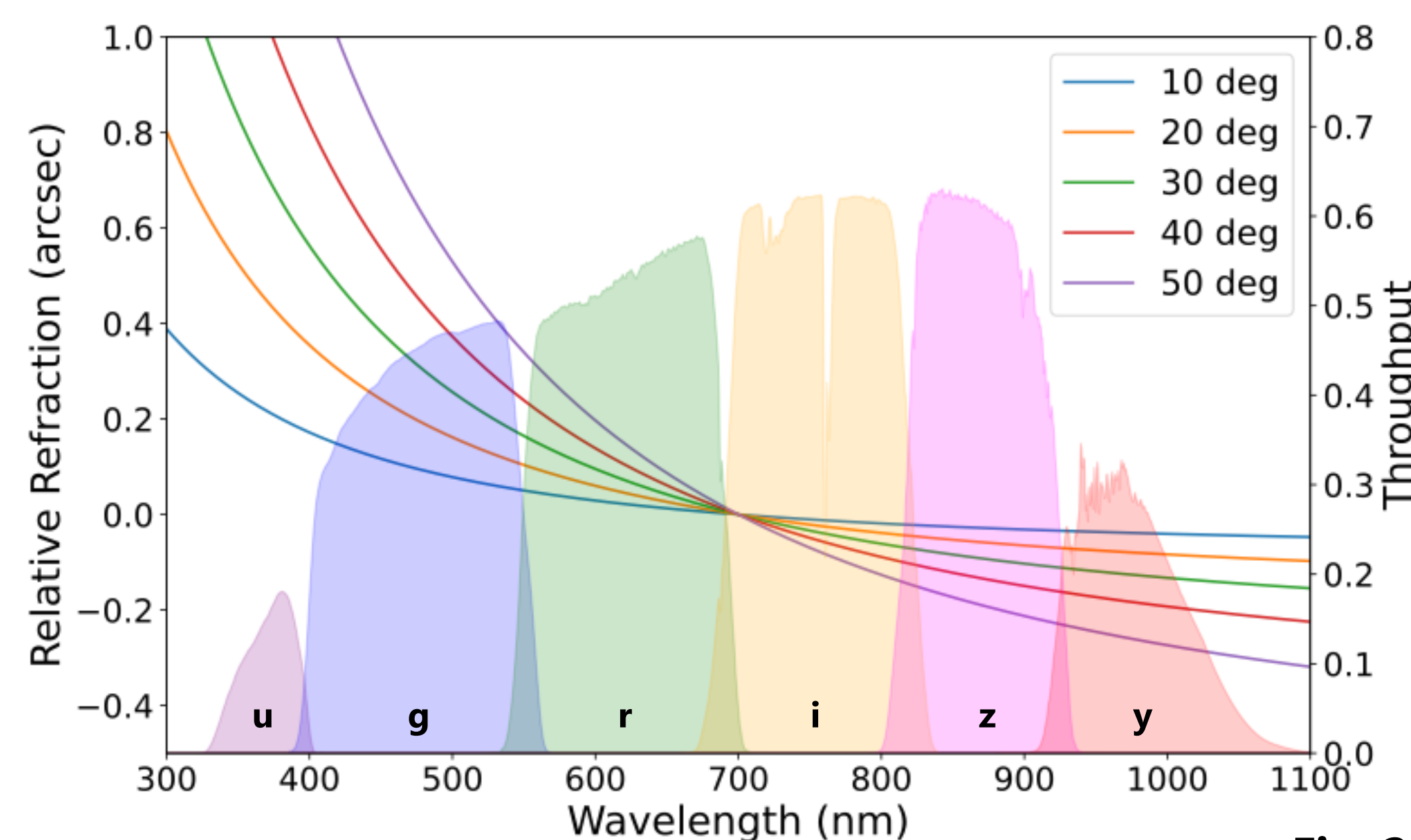
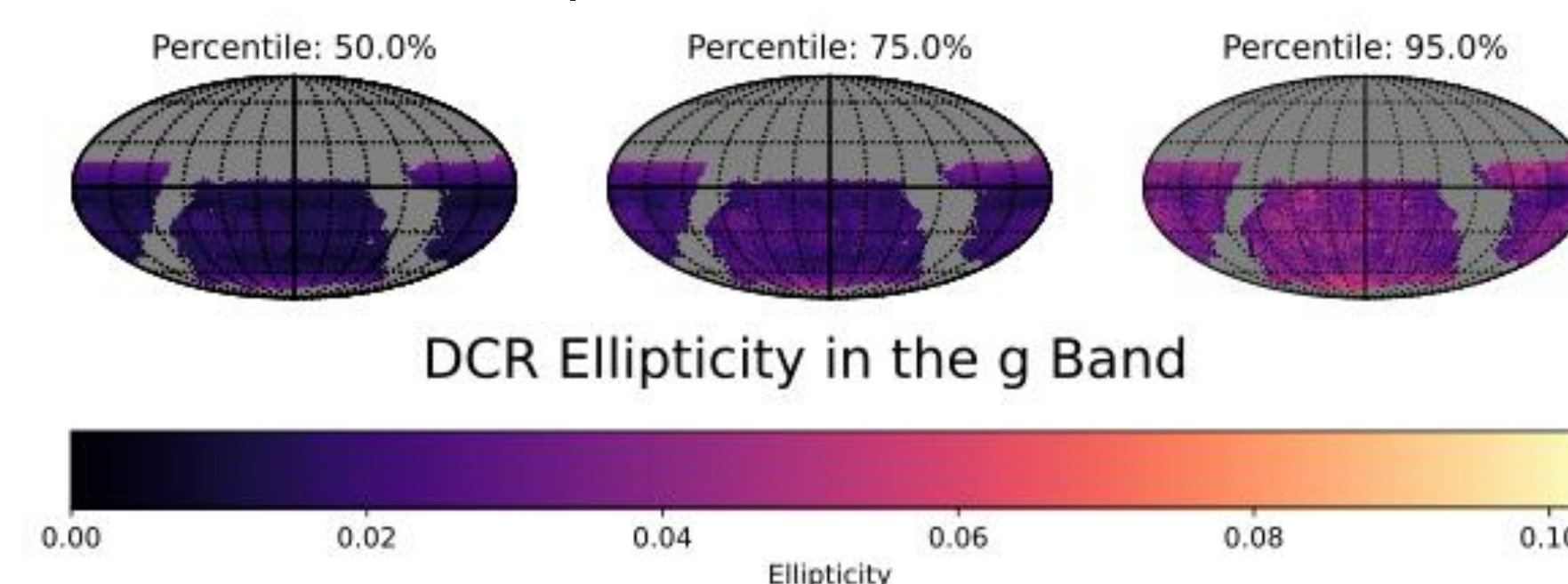


Fig. 2

Relative refraction angle for five different zenith angles (solid curves, left axis) and response curves in each LSST band (filled regions, right axis) as a function wavelength.

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